- The greatest common divisor of two positive integers a and b ($gcd(a,b)$) can be determined by a procedure known as Euclid's Algorithm. It is based on the theorem that

$$gcd(a,b) = gcd(b, a \bmod b)$$

- The expression 'mod' is used in moduler arithmetic which is a special kind of arithmetic involving remainders as will be seen next.

Moduler Arithmetic

- The symbol used ($\equiv$) is known as the congruence symbol.

- Moduler relationships are of the form $n \equiv R \pmod{m}$ (spoken as "n is congruent to R mod m") where n and R are integers and m is a positive integer known as the modulus.

- If this congruence relationship holds, then it is said that n is congruent to R modulo m.

- The modulus operator (mod) produces the remainder when the integer on it's left is divided by the modulus.

- Thus the term (R mod m) is equal to the remainder r, when R is divided by m.

- If two remainders are equal then it can be written that $(n \bmod m) \equiv (R \bmod m)$ - a standard equality.

- However, if the modulus is equal on bothsides of the equation, then the (mod m) term can be removed from the left side and the equality symbol replaced with a congruence symbol (along with a slight re-arrangement of the brackets).

- Assuming $n \neq r$, it would be <u>incorrect</u> to say $n = (R \bmod m)$.

- However, it is correct to say that $n \equiv R \pmod{m}$ and this bassically states that the same remainder (in this case r) results when both n and R are divided by m.

- As mentioned, the remainder r is also known as a <u>residue</u>.

- If $R = r$ (i.e $0 \le R < m$) then R is known as a <u>least residue</u>

- The congruent relation '$\equiv$' is an equivalence relation.

- The set of numbers congruent to some value 'a' $(\bmod\ m)$ is known as a <u>residue class</u> (or a congruent class)

- As $0 \le r < m$, this means there are m possible values of r and hence there are m possible residue classes.

- The congruence relationship $n \equiv R \pmod{m}$ is only true if $m \mid n-R$

- If $a \equiv a_1 \pmod{m}$ and $b \equiv b_1 \pmod{m}$, then $a+b \equiv a_1+b_1 \pmod{m}$ and $ab \equiv a_1 b_1 \pmod{m}$

- The integers modulo m, denoted Z_m is the set of integers $\{0,1,2,\cdots,m-1\}$

- To understand why, it must be remembered that the integers n and R can be expressed as $q_{\{n,R\}} \times m + r_{\{n,R\}}$ where the subscript $\{n,R\}$ represents the fact that q and r will generally take on different values for n and R.

- Only if $r_n = r_R$ will $m \mid (n-R)$ because in this case the two remainders cancel each other in the $(n-R)$ term:

$$(q_n \times m + r_n - q_R \times m - r_R) = (q_n \times m - q_R \times m)$$

- Because $m \mid (q_n \times m)$ and $m \mid (q_R \times m)$ $\Rightarrow m \mid (q_n \times m - q_R \times m)$. If $(n-R)$ is not divisible by m then the notation used to represent this is $\not|$ and therefore $m \not| (n-R)$. In this case $n \not\equiv R \pmod{m}$.

Moduler Inverse

- The idea of an inverse is important both in ordinary arithmetic and moduler arithmetic.

- In any set of numbers, the inverse of a number contained in that set is another number which when combined with the first under a particular operation will give the Identity element for that operation.

- The identity element for a particular operation is a number that will leave the original number unchanged under that operation.

- Two example of inverses are the: (1) Additive inverse, (2) Multiplicative inverse.

- The Identity element under different operations will be different.

- Under addition, it is '0', as any number added to '0' will remain unchanged.

- However, under normal multiplication the identity element is 1 as any number multiplied by 1 will remain unchanged.

- In ordinary arithmetic if the number is x then the additive inverse is $-x$ and multiplicative inverse is $\frac{1}{x}$.

- The idea is same in moduler arithmetic however if x is an integer then its multiplicative inverse would not be $\frac{1}{x}$ as there is no such things as a fraction in modular arithmetic.

- In this case, it would be a number which, when multiplied by the original number, would give a result that is congruent 1 (mod m) (again, m is the modulus).

- A number x can only have a multiplicative inverse if it is relatively prime to the modulus (i.e $\gcd(x, m) = 1$)

- When one number is operated on modulo some other number, it is said that the first number has been reduced modulo the second and the operation is called a modular reduction.

- Let $a \in Z_m$, the multiplicative inverse of a modulo m is an integer x belongs to Z_m such that $ax \equiv 1 \pmod{m}$. If such an 'x' exists, then it is unique and a is said to be invertible. The inverse of a is denoted by a^{-1}.

Abstract Algebra

- Will only be looking at a very small subset of what this subject has to offer.

- Three main ideas here that need to be grasped:
  1. Group $(G, \cdot)$
  2. Ring $(Rg, +, \times)$
  3. Field $(F, +, \times)$

- Basically, three different types of sets along with some operations.

- The classification of each set is determined by the axioms which is satisfies.

Group

- A Group $\{G, \cdot\}$ is a set under some operation $(\cdot)$ if it satisfies the following 4 axioms:

  1. closure (A_1): For any two elements $a, b \in G$, $c = a \cdot b \in G$.

2. Associativity (A_2) : For any three elements $a, b, c \in G$,

$$(a \cdot b) \cdot c = a \cdot (b \cdot c)$$

3. Identity (A_3) : There exists an Identity element $e \in G$ such that $\forall a \in G$, $a \cdot e = e \cdot a = a$.

4. Inverse (A_4) : Each element e in G has an inverse i.e $\forall a \in G$ $\exists a^{-1} \in G$, $a \cdot a^{-1} = a^{-1} \cdot a = e$

- However it is said to be an Abelian group if in addition to the above the set follows the axiom :

5. Commutativity (A_5): For any $a, b \in G$
$$a \cdot b = b \cdot a$$

Cyclic group

- Exponentiation is the repeated application of the group operator.